Please check that this question paper contains 09 questions and 02 printed pages within first ten minutes.

[Total No. of Questions: 09] [Total No. of Pages: 02]

Uni. Roll No. ....

Program: B.Tech. (Batch 2018 onward)
Semester: 3
Name of Subject: Mathematics-III
Subject Code: BSCS-101
Paper ID: 16013
Scientific calculator is Allowed

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07 JUN 2023

Time Allowed: 03 Hours **Max. Marks: 60**

NOTE:

1) Parts A and B are compulsory
2) Part-C has Two Questions Q8 and Q9. Both are compulsory, but with internal choice
3) Any missing data may be assumed appropriately

Part – A **[Marks: 02 each]**

Q1

a) Separate $\sin z$ into real and imaginary parts.

b) Discuss Gauss Jordan method for solving three equations in x, y, z.

c) Write algorithm for straight line fit of a curve.

d) Find the mean and variance of Binomial distribution when $n = 12$ and $p = \frac{1}{3}$.

e) Evaluate $\oint_C \frac{e^{-z}}{z+1} dz$, where C is the circle $|z| = 2$.

f) A sample of 400 male students is found to have a mean height 67.47 inches. Can it be reasonably regarded as sample from a large population with mean height 67.39 inches and standard deviation 1.30 inches. (tabulated value=1.96)

Part – B **[Marks: 04 each]**

Q2. Determine a, b, c, d so that the function $f(z) = (x^2 + axy + by^2) + i(cx^2 + dxy + y^2)$ is analytic.

Q3. Fit a straight line to the following data:

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x	1	2	3	4	5
y	5	7	9	10	11

Q4. Solve the system of equations $x+2y+z=3,\ 2x+3y+3z=10,\ 3x-y+2z=13$ using Gauss elimination method.

Q5. Find the Laurent's series of $f(z)=\dfrac{1}{(1-z)(z-2)}$ for the following intervals

(a) $1<|z|<2$ (b) $|z|>2$

Q6. If the probability that an individual suffers a bad reaction from a certain injection is 0.001. Find the probability that out of 2000 individuals

(a) exactly 3 individuals will suffer a bad reaction

(b) none will suffer a bad reaction.

(c) more than one individual will suffer.

Q7. A random sample of size 16 has 53 as mean. The sum of square of deviation from mean is 135. Can this be regarded as taken from the population having 56 as mean? (tabulated value =2.731)

Part – C **[Marks: 12 each(6 for each subpart if any)]**

Q8. Evaluate the integral $\int_0^{2\pi}\dfrac{d\theta}{5-3\cos\theta}$ using contour integration.

OR

(i) Determine the analytic function $f(z)$ in terms of z whose real part is $e^x(x\cos y - y\sin y)$

(ii) Find the bilinear transformation which maps the points $z=1,-i,-1$ to the Points $w=i,0,-i$.

Q9. Fit a Binomial distribution to the following data and test for goodness of fit:

x	0	1	2	3	4	5
f	38	144	342	287	164	165

(tabulated value=11.07).

OR

Suppose that the p.d.f of a random variable X is as follows:

$$f(x)=\begin{cases} cx & for\ 0<x<4 \\ 0 & elsewhere \end{cases}$$

where c is a given constant. Determine the value of c and the values of $P(1\le X\le 2)$ and $P(X>2)$.

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